

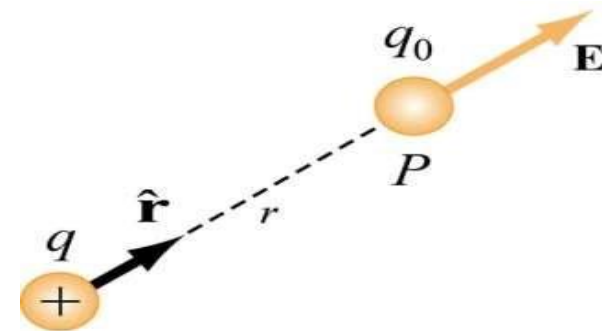
Electric field and Intensity

Coulomb's Law suggests an *instantaneous interaction*: no matter how far apart the charges are, they instantly “know” if the location of another charge has been changed (“action-at-a-distance”)

But In case of electrostatics field, Instead of “action-at-a-distance” between the charges, we can consider *the interaction of a charge with the field created by all other charges at its location.*

To find the electric field due to a *point charge*(q) we place a *positive test charge* (q_0) at any point near the particle, at distance r . From Coulomb's law the force on the test charge due to the particle with charge q is

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^2} \hat{r}.$$



The magnitude of electric field intensity at any point in electric field is given by force that would be experienced by a unit positive test charge placed at that point.

Then it can be written as,

$$\vec{E} = \frac{\vec{F}}{q_0} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

In Magnitude form,

$$E = \frac{1}{4\pi\epsilon_0} \frac{|q|}{r^2}$$

The unit of Electric field (intensity) is N/C or V/m.

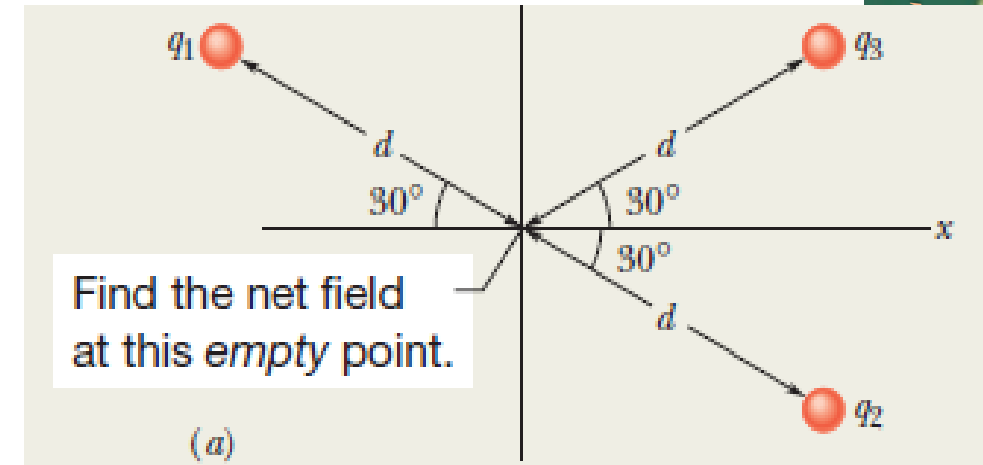
If several electric fields are set up at a given point by several charged particles, forces obey the principle of superposition, so we just add the forces as vectors,

$$\begin{aligned} \vec{F}_0 &= \vec{F}_{01} + \vec{F}_{02} + \dots + \vec{F}_{0n} \quad \Rightarrow \quad \vec{E} = \frac{\vec{F}_0}{q_0} = \frac{\vec{F}_{01}}{q_0} + \frac{\vec{F}_{02}}{q_0} + \dots + \frac{\vec{F}_{0n}}{q_0} \\ \vec{F}_{net} &= \sum_i \vec{F}_i \quad \longleftrightarrow \quad \vec{E}(\vec{r}) = \sum_i \vec{E}_i(\vec{r}) \quad = \vec{E}_1 + \vec{E}_2 + \dots + \vec{E}_n \end{aligned}$$

This tells us that electric fields also obey the principle of superposition.

Self Test (635 E)

Figure shows three particles with charges $q_1 = +2Q$, $q_2 = -2Q$, and $q_3 = -4Q$, each a distance d from the origin. What net electric field $E \rightarrow$ is produced at the origin?



$$\frac{6.93Q}{4\pi\epsilon_0 d^2}$$



Electric Potential and Electric Potential Energy

We define *change in electric potential energy of the system* as negative of the work done by the electric forces.

$$W = -\Delta U \quad (\text{work, conservative force}).$$

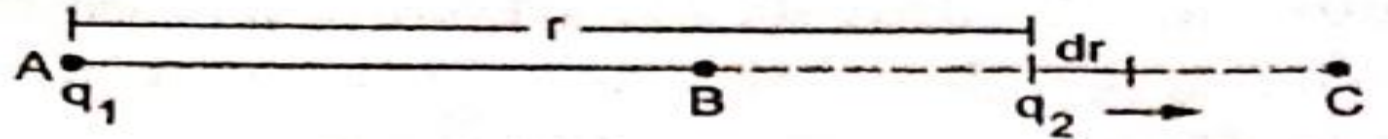
For particle moving from initial position i to final position f, work done can be written as

$$W = -\Delta U = -q \Delta V = -q(V_f - V_i).$$

For the two system charges q_1 and q_2 as shown in figure

Taking .

$AB = r_1$ and $AC = r_2$



‘ r ’ is the position of test charge q_2 from point A which varies as $r_1 < r < r_2$.

For Small displacement of charge q_2 from r to $r + dr$, the electric force is

$$F = \frac{q_1 q_2}{4\pi\epsilon_0 r^2} \text{ towards } \vec{AB}.$$

Small work done by this force in the small displacement dr is

$$dW = \frac{q_1 q_2}{4\pi\epsilon_0 r^2} dr.$$

The total work done as the charge q_2 moves from B to C is

$$W = \int_{r_1}^{r_2} \frac{q_1 q_2}{4\pi\epsilon_0 r^2} dr = \frac{q_1 q_2}{4\pi\epsilon_0} \left(\frac{1}{r_1} - \frac{1}{r_2} \right).$$

Change in potential energy

$$U(r_2) - U(r_1) = -W = \frac{q_1 q_2}{4\pi\epsilon_0} \left(\frac{1}{r_2} - \frac{1}{r_1} \right).$$

we choose Potential energy of the two charge system is Zero when they have infinite separation i.e. $r_1 = \text{infinity}$ [$U(r_1) = 0$]. When the separation is r ,

Then,

$$U(r) = U(r) - U(\infty) \\ = \frac{q_1 q_2}{4\pi\epsilon_0} \left(\frac{1}{r} - \frac{1}{\infty} \right) = \frac{q_1 q_2}{4\pi\epsilon_0 r} .$$

This equation is derived assuming that one of the charge (q_1) is fixed and other (q_2) is displaced.

Electric Potential at Point:

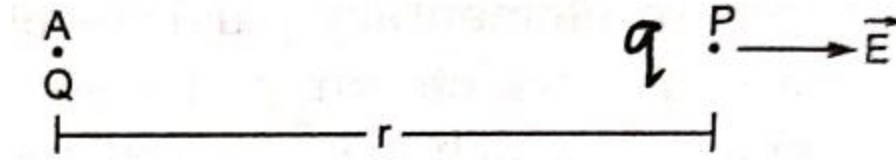
Change in Electric potential energy per unit test charge when it is moved from reference point (say) 'P' to a point (say) 'A' is called electric potential at that point 'A'.

$$V_A = V_A - V_P = \frac{U_A - U_P}{q} .$$

Electric field due to point charge.

From figure , Coulomb's law

$$F = \frac{Qq}{4\pi\epsilon_0 r^2}$$



Then electric field due to a point charge is, By definition

$$E = \frac{F}{q} = \frac{Q}{4\pi\epsilon_0 r^2}$$

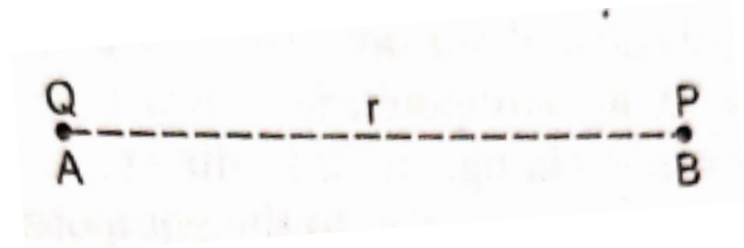
Electric potential due to point charge

Taking reference point at $r = \text{infinity}$. Suppose a test charge q is brought from $r = \text{infinity}$ to the point P. The change in electric potential energy of the system is

$$U_P - U_\infty = \frac{Qq}{4\pi\epsilon_0 r}.$$

From Definition,

$$V_P = \frac{U_P - U_\infty}{q} = \frac{Q}{4\pi\epsilon_0 r}.$$



(In general)

For the system of charges,

$$V = \frac{1}{4\pi\epsilon_0} \sum \frac{Q_i}{r_i}.$$

Relation between Electric field and potential

Suppose a point charge q is displaced slightly from the point $\mathbf{r}$ to $\mathbf{r} + d\mathbf{r}$, the force on the charge is

$$\mathbf{F} = q\mathbf{E}$$

Work done by the field is

$$dW = \mathbf{F} \cdot d\mathbf{r} = q\mathbf{E} \cdot d\mathbf{r}$$

Change in potential energy is

$$dU = -dW = -q\mathbf{E} \cdot d\mathbf{r}$$

Change in potential

$$dV = dU/q \Rightarrow dV = -\mathbf{E} \cdot d\mathbf{r}$$

Integrating between the points $\mathbf{r}_1$ and $\mathbf{r}_2$ we get,

$$V_2 - V_1 = - \int_{\mathbf{r}_1}^{\mathbf{r}_2} \vec{E} \cdot d\vec{r}$$

Where V_1 and V_2 are the potentials at $\mathbf{r}_1$ and $\mathbf{r}_2$ resp.

Choosing $\mathbf{r}_1$ as reference point (at infinity) and $\mathbf{r}_2$ at $\mathbf{r}$ then above equation becomes

BOLD LETTERS INDICATE VECTOR QUANTITY

$$V(\vec{r}) = - \int_{\infty}^{\vec{r}} \vec{E} \cdot d\vec{r}$$

This is the relation between electric potential and field.

Electric field due to Dipole

An electric dipole consists of two particles with charges of equal magnitude q but opposite signs, separated by a small distance d . The electric dipole moment (vector) $\mathbf{p}$ has magnitude qd and points *from the negative charge to the positive charge*. The unit of $\mathbf{p}$ is the coulomb-meter (c-m).

At Axial line:

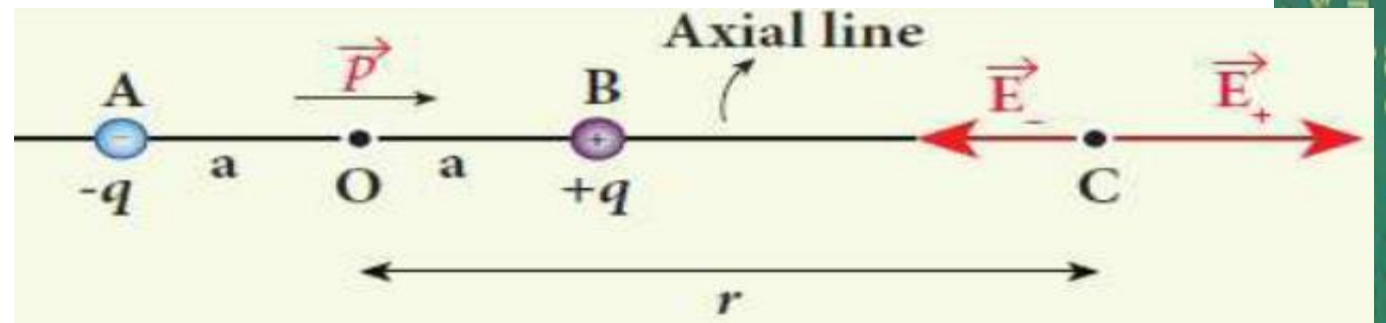
Electric field due to $+q$ at C is

$$\vec{E}_+ = \frac{1}{4\pi\epsilon_0} \frac{q}{(r-a)^2}$$

Similarly due to $-q$ at C is

$$\vec{E}_- = -\frac{1}{4\pi\epsilon_0} \frac{q}{(r+a)^2}$$

From this figure, $E_+ > E_-$ and dipole moment is $\mathbf{p} = 2qa$.



- Total Electric field at point C is

$$\vec{E}_{tot} = \vec{E}_+ + \vec{E}_- \\ = \frac{1}{4\pi\epsilon_0} \frac{q}{(r-a)^2} - \frac{1}{4\pi\epsilon_0} \frac{q}{(r+a)^2}$$

$$\vec{E}_{tot} = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{(r-a)^2} - \frac{1}{(r+a)^2} \right) \quad \text{Along } E_+$$

If point C is far away from the dipole ($r \gg a$) then $(r^2 - a^2)^2 \approx r^4$

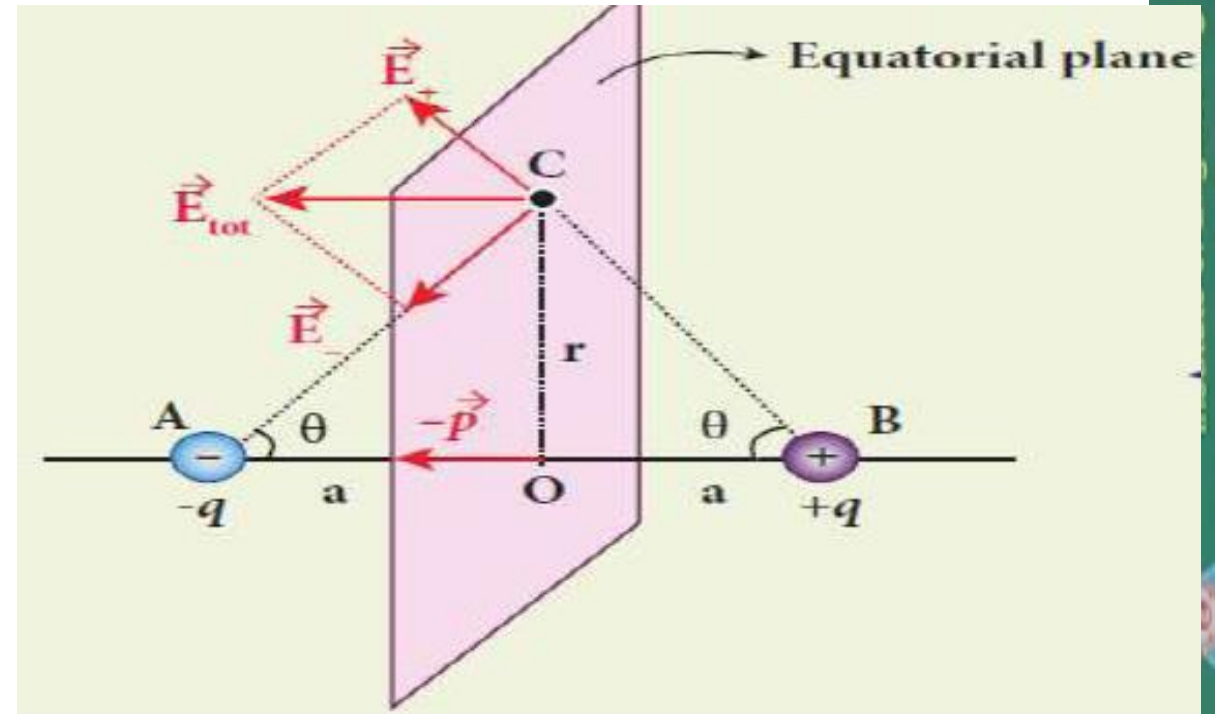
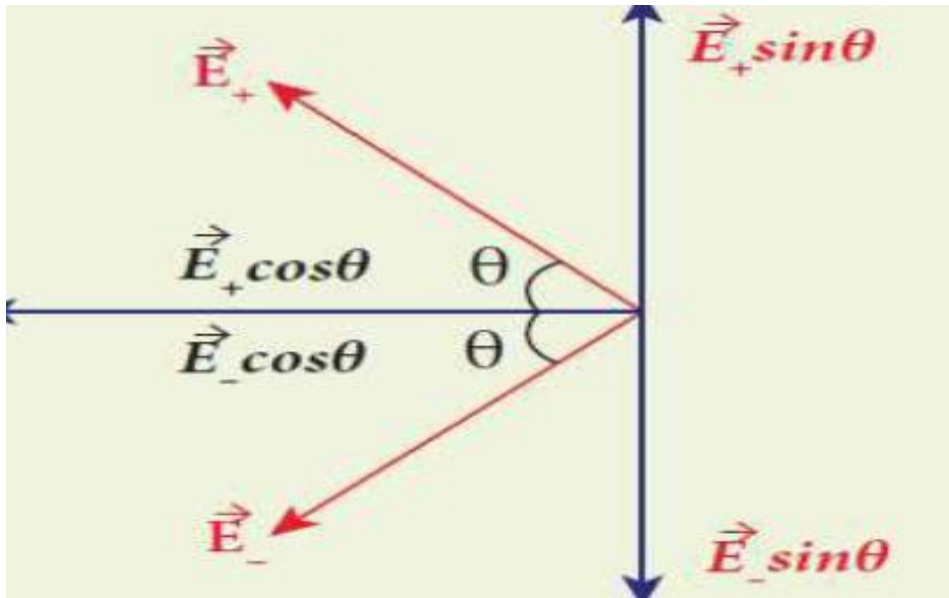
$$\vec{E}_{tot} = \frac{1}{4\pi\epsilon_0} \left(\frac{4aq}{r^3} \right)$$

Taking $2qa = \mathbf{p}$ as dipole moment then,

$$\vec{E}_{tot} = \frac{1}{4\pi\epsilon_0} \frac{2\vec{p}}{r^3} \quad (r \gg a)$$

At Equatorial line:

The magnitude of E_+ and E_- at C are equal and the directions are as shown in figure. The perpendicular component $E_+ \sin\theta$ and $E_- \sin\theta$ are oppositely directed and cancel each other.



Total electric field at point C is given by

$$E_{\text{tot}} = 2E_+ \cos\theta \quad (\text{Since magnitude of } E_+ = E_- = \frac{1}{4\pi\epsilon_0} \frac{q}{(r^2 + a^2)}).$$

$$= \frac{1}{4\pi\epsilon_0} \frac{2q \cos\theta}{(r^2 + a^2)}$$

$$\text{since } \cos\theta = \frac{a}{\sqrt{r^2 + a^2}}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{2qa}{(r^2 + a^2)^{\frac{3}{2}}}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{\vec{p}}{(r^2 + a^2)^{\frac{3}{2}}} \quad \mathbf{p} = 2qa \text{ dipole moment,}$$

At very large distance ($r \gg a$), $E_{\text{tot}} =$

$$\frac{1}{4\pi\epsilon_0} \frac{\vec{p}}{r^3}$$

along the direction opposite to that of

Conclusions:

- It is inferred that for very large distances, the magnitude of the electric field at points on the dipole axis is twice the magnitude of the electric field at points on the equatorial plane.
- The direction of the electric field at points on the dipole axis is directed along the direction of dipole moment vector p but at points on the equatorial plane it is directed opposite to the dipole moment vector, that is along $-p$.
- At very large distances, the electric field due to a dipole varies as $1/r^3$. Note that for a point charge, the electric field varies as $1/r^2$. This implies that the electric field due to a dipole at very large distances goes to zero faster than the electric field due to a point charge.
- The reason for this behavior is that at very large distance, the two charges appear to be close to each other and neutralize each other.

Electric potential due to dipole